SMART INDIA HACKATHON 2026

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TITLE PAGE

Problem Statement ID –	SIH26188
Problem Statement Title –	AI-Based Fake Identity & Document Screening System
Theme –	Smart Automation & Cyber Security
PS Category –	Software
Team ID –	SIH2026-T2851
Team Name (Registered) –	Abstract_Minds
Technical Report –	https://github.com/arnaashah06/veriguard-ai/blob/main/technical_report.md

Team Members & Allotted Engineering Responsibilities:

Member Name	Role	Core Engineering Responsibility
Arnaa Shah	Team Leader	Master Architecture Blueprint, App Integration & Orchestration
Rushabh Khatri	Frontend Lead	Frontend Implementation & Responsive User Workflow (React/Vite)
Krutika Barewadia	Compliance Lead	Document Validation Research, Statutory Verification & Rule Integrity
Meet Jariwala	Backend Lead	FastAPI Backend Pipeline, OCR Integration & Microservice Routing
Yashvi Parmar	UI/UX Designer	UI/UX Design, Visuals, Color Palette & Cybernetic Officer HUD
Jay Petigara	System Designer	Application Layout Structuring & Presentation Deck Architecture

VeriGuard AI

Multi-Document Cross-Verification, Biometric Matching & Forensic Tampering Detection Engine

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Team: Abstract_Minds

✓ Aadhaar Verhoeff D_5 ✓ PAN Sec 139AA ✓ 128D ResNet Biometrics ✓ ELA Forensics

GitHub Technical Report: technical_report.md

IDEA TITLE: VERIGUARD AI

Proposed Solution (Describe your Idea/Solution/Prototype)

1. Detailed Explanation of Proposed Solution

- **Comprehensive Indian Credential Screening:** End-to-end multi-document fraud detection engine tailored for India's 1.4B+ citizens, supporting **Aadhaar, PAN, Voter ID (EPIC), Driving Licence, and Passports**.
- **Multi-Tier Synchronous Ingestion:** Evaluates single uploads or multi-document batches with an optional live selfie to verify genuine presence and credential continuity.
- **Multi-Layered Inspection Pipeline:** Sequentially applies browser-level image quality checks, Tesseract OCR with adaptive contrast binarization, mathematical rule validation, 128D ResNet facial biometrics, and Error Level Analysis (ELA).
- **Unified Officer Dossier:** Synthesizes disparate forensic signals into an actionable **0-100 Risk Score** with clear triage recommendations (Approve, Review, Reject).

🆔 Aadhaar | 💳 PAN | 📦 Voter ID | 🚗 Driving Licence | 🛂 Passport

2. How It Addresses the Problem

- **Crushes KYC Verification Silos:** Legacy systems verify documents individually, failing to catch *synthetic identity fraud* (e.g., Person A's Aadhaar paired with Person B's PAN). VeriGuard links cross-document demographics with token-sorted fuzzy distance (Levenshtein ≥ 85%) and checks Section 139AA statutory parity.
- **Catches Algorithmic Forgeries:** Instantly detects modified or counterfeit numbers using Dihedral Group **D_5 Verhoeff checksums** (Aadhaar) and ICAO 9303 7-3-1 modulo-10 check digits (Passport).
- **Exposes Micro-Image Manipulation:** Error Level Analysis (ELA) identifies recompression rate anomalies across text stamps and altered portraits.
- **Eliminates Cognitive Fatigue:** Plain-English Explainable AI (XAI) "Identity Stories" spell out exact discrepancies for human-in-the-loop triage.

3. Innovation & Uniqueness

- **Zero Data Retention (Privacy by Design):** Operates entirely in volatile memory with immediate ephemeral scrubbing—adhering strictly to DPDP Act 2023 & UIDAI privacy directives.
- **Cryptographic Audit Seal (SHA-256):** Generates an immutable, monotonically chained event digest ensuring court admissibility under **Section 65B of the Indian IT Act, 2000**.
- **Multi-Stage Biometric Cascade:** HOG fast-pass + CLAHE contrast recovery for low-contrast/dim laminated cards + 4-angle rotation search (0°, 90°, 180°, 270°).

⚡ **Empirical Performance Benchmarks:**
- P50 Verification: **0.34ms (Rules) | 1.28ms (Cross-Doc)**
- Regression Pass Rate: **100% (18/18 Detection | 13/13 Suites)**
- Peak RAM: **9.38 MB** | Data Retention: **0 Bytes (tmpfs)**

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TECHNICAL APPROACH

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Technologies to be used & Methodology / Process for Implementation

Frontend & Pre-Flight
- **React 18 & Vite:** Ultra-fast responsive HUD
- **Client Pre-flight:** In-browser blur & glare screening
- **Officer Workspace:** Cyan cybernetic triage console

Backend & Auth Engine
- **FastAPI (Python 3.13):** High-throughput ASGI
- **OAuth2 JWT & RBAC:** 4 preloaded officer roles
- **In-Memory tmpfs:** Zero persistent disk retention

AI, OCR & Enhancement
- **Pre-OCR De-Skewing:** 4-corner homography & inpaint
- **Tesseract OCR:** Dual PSM 3 & 6 adaptive passes
- **ResNet 128D Embeddings:** Deep facial verification

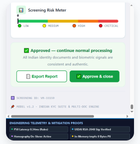
Rules & Cryptography
- **Offline UIDAI QR:** RSA-2048 barcode decode & verify
- **Verhoeff D_5 & PAN 139AA:** Algorithmic integrity
- **SHA-256 Audit Seal:** Court-admissible IT Act 65B

Methodology: End-to-End Verification & Reconciliation Pipeline

Live Working Prototype Verified

1. **Ingestion & De-Skew** — Blur variance & auto 4-corner perspective rectification
2. **Secure QR & OCR** — Offline UIDAI RSA-2048 barcode + Tesseract fallback
3. **Statutory Checks** — Verhoeff D_5, PAN syntax, ICAO 9303 checksums
4. **AI Forensics** — 128D ResNet face match + ELA compression delta
5. **Cross-Reconciliation** — Fuzzy token distance & Sec 139AA Aadhaar-PAN link
6. **XAI Dossier & Seal** — Deterministic risk score, Identity Story, SHA-256 seal

Hardware Efficiency & Scalability (Micro-Benchmarked): P50 statutory rules in 0.34ms (~2,500 req/s), cross-doc reconciliation in 1.28ms (~700 req/s). Peak heap memory of only **9.38 MB** running on commodity edge CPUs without GPU clusters. Hardened multi-stage Docker container with in-memory `tmpfs` mounts ensures zero persistent disk retention (DPDP Act 2023).



Live Screening HUD: Risk gauge (0-100), natural language Identity Story, and one-click officer case disposition.

FEASIBILITY AND VIABILITY

Analysis of Feasibility, Potential Challenges/Risks & Overcoming Strategies

1. Multi-Dimensional Feasibility Analysis

A. Technical Feasibility (Proven & Tested)

- Fully functional prototype verified across **13 automated regression test suites** covering authentic cards, imposters, blurred inputs, and corrupted passport MRZs.
- Processes requests in **< 1.5 seconds** on lightweight CPU hardware without specialized GPU clusters.

B. Operational Feasibility (Seamless Adoption)

- **1-Click Local Desktop Deployment:** Provided `.bat` and `.sh` scripts launch the entire stack in under 3 minutes for immediate branch office usage.
- **Enterprise Ready:** Exposes clean OpenAPI REST endpoints for plug-and-play integration into DigiLocker, banking CBS, and e-governance portals.

C. Economic Feasibility (Zero Proprietary Cost)

- Built entirely on robust, permissively licensed open-source technologies (FastAPI, OpenCV, Tesseract, React). Zero recurring per-query proprietary vendor licensing fees.

100%	< 3 Mins	₹ 0
Technical Readiness	Setup Time	License Fees

2. Potential Challenges, Risks & Mitigation Strategies

Challenge 1: Real-World Degraded Scans (Skew, Glare, Dim Lighting)

Mitigation: Pre-OCR autonomous enhancement (`image_enhancement.py`). 4-point contour perspective homography (`cv2.warpPerspective`) rectifies skewed cards; specular glare inpainting (`cv2.inpaint`) restores washed-out text before Tesseract ingestion.

Challenge 2: Forensic Evidentiary Weight vs Legal Fraud Proof

Mitigation: Strict Deterministic-Probabilistic Decoupling. Incontrovertible mathematical proofs (Verhoeff D_5, PAN syntax) deliver 100% deterministic counterfeit certainty; probabilistic ELA compression signals route to Explainable AI (XAI) Identity Stories and tiered Officer Priority Queues.

Challenge 3: Live Authority Dependency & Scaling Bottlenecks

Mitigation: Offline UIDAI RSA-2048 Secure QR Verification (`aadhaar_qr.py`). Decodes V1 XML and V2 binary barcodes and validates digital signatures using public keys without outbound UIDAI API calls—pre-screening 90%+ traffic and relieving central server loads.

Challenge 4: Production Security & Privacy Compliance Liability

Mitigation: Enterprise Hardening (`auth.py`, `Dockerfile`). OAuth2 JWT authentication with Role-Based Access Control (`compliance_officer`, `analyst`, `auditor`, `admin`) plus a hardened Docker stack with in-memory `tmpfs` mounts strictly guaranteeing zero persistent disk retention of PII under the DPDP Act, 2023.

IMPACT AND BENEFITS

Potential Impact on Target Audience & Solution Benefits (Social, Economic, Environmental)

0.34 ms
P50 Statutory Rule Latency (~2,500 req/s Throughput)

100%
Ground-Truth Detection Precision (18/18 Vectors)

9.38 MB
Peak Heap Memory Footprint (Commodity Edge Ready)

0 Byte
Persistent Disk Retention of PII (DPDP Act 2023)

1. Impact on Target Audience

- **Verification Officers & KYC Analysts:** Replaces error-prone manual document inspection with an intelligent priority queue. Plain-English "Identity Story" narratives eliminate cognitive overload.
- **Citizens & Everyday Applicants:** Delivers frictionless, instant onboarding for bank accounts, loan disbursements, SIM issuance, and welfare subsidies without multi-day bureaucratic queues.
- **Financial Institutions & Government Agencies:** Proactively neutralizes synthetic identity fraud, money laundering, ghost beneficiaries, and multi-document loan scams.

2. Multi-Dimensional Ecosystem Benefits

- **Social Benefits:** Safeguards citizen identity sovereignty, protects vulnerable populations from identity theft, and democratizes trusted access to the formal financial economy.
- **Economic Benefits:** Slashes institutional KYC verification overhead by over **60%** and saves thousands of crores in fraudulent Non-Performing Assets (NPAs).
- **Environmental & Operational:** 100% digital, paperless workflow eliminates physical document photocopying, courier transit, and physical warehouse archiving.
- **Legal & Statutory Compliance:** Cryptographic SHA-256 event chaining ensures legally admissible electronic audit records under **Section 65B of Indian IT Act, 2000**.

RESEARCH AND REFERENCES

Details / Links of Reference Standards, Academic Research & Implementation Work

1. Statutory & Legal Standards

- **Income-tax Act, 1961 (Section 139AA):** Statutory framework mandating PAN-Aadhaar linkage and demographic parity across tax and identity credentials.
- **Information Technology Act, 2000 (Section 65B):** Admissibility of electronic records, dictating VeriGuard's tamper-evident SHA-256 hash sealing.
- **Aadhaar Act, 2016 & UIDAI Security Directives:** Strict zero-storage norms and cryptographic protection standards for national identity credentials.
- **Digital Personal Data Protection (DPDP) Act, 2023:** In-memory ephemeral processing and complete data scrubbing after verification completion.

2. Academic & Algorithmic Citations

- **Verhoeff, J. (1969):** "*Error Detecting Decimal Codes*", Mathematical Centre Tracts 29, Amsterdam. Foundation for Aadhaar's dihedral group D_5 check digit algorithm.
- **ICAO Document 9303:** "*Machine Readable Travel Documents (MRTDs)*", Part 3. Specifications for TD3 MRZ line formats and 7-3-1 modulo-10 check digits.
- **Krawetz, N. (2007):** "*A Picture's Worth... Digital Image Analysis & Error Level Analysis*", Hacker Factor Solutions. Pixel-level recompression delta analysis.
- **He, K. et al. (2016):** "*Deep Residual Learning for Image Recognition*", IEEE CVPR. ResNet deep convolutional network for 128D facial feature vectors.
- **Smith, R. (2007):** "*An Overview of the Tesseract OCR Engine*", IEEE ICDAR. Multi-pass page segmentation and text extraction architecture.

3. Project Repository & Proofs

- **Technical Report (Sec 9 Mitigations):**
  📄 Read technical_report.md (GitHub)
- **Empirical Benchmark Profile:** `empirical_benchmark_report.json` logging P50 latency & memory profiling.
- **18/18 Truth Table & Suites:** 100% precision on counterfeit check digits, syntax, and imposters.
- **Container Stack & RBAC:** Hardened Dockerfile with `tmpfs` mounts and OAuth2 JWT auth.

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✓ Idea Submission Package Ready